

E - Count 123

分值: 450 分

Problem Statement

Find the number, modulo 998244353, of sequences $A = (a_1, \dots, a_{X_1+X_2+X_3})$ of length $X_1 + X_2 + X_3$ that satisfy all of the following conditions.

求满足以下全部条件、长度为 $X_1 + X_2 + X_3$ 的序列 $A = (a_1, \dots, a_{X_1+X_2+X_3})$ 的数量, 答案对 998244353 取模。

- A contains exactly X_1 copies of 1, exactly X_2 copies of 2, and exactly X_3 copies of 3.
- The absolute difference between adjacent elements is at most 1. That is, for every integer i satisfying $1 \leq i \leq X_1 + X_2 + X_3 - 1$, we have $|a_{i+1} - a_i| \leq 1$.
- A 中恰好包含 X_1 个 1、 X_2 个 2, 以及 X_3 个 3。
- 相邻元素的绝对值之差不超过 1。即, 对于所有满足 $1 \leq i \leq X_1 + X_2 + X_3 - 1$ 的整数 i , 都有 $|a_{i+1} - a_i| \leq 1$ 。

Constraints

- $1 \leq X_1, X_2, X_3 \leq 10^6$
- All input values are integers.
- $1 \leq X_1, X_2, X_3 \leq 10^6$
- 所有输入值都是整数。

Input

The input is given from Standard Input in the following format:

输入按照以下格式从标准输入给出:

$X_1 \ X_2 \ X_3$

Output

Output the answer.

输出答案。

Sample Input 1

2 2 1

Sample Output 1

9

The sequences satisfying the conditions are the following nine:

满足条件的序列共有以下 9 个：

- (1, 1, 2, 2, 3)
- (1, 1, 2, 3, 2)
- (1, 2, 1, 2, 3)
- (1, 2, 3, 2, 1)
- (2, 1, 1, 2, 3)
- (2, 3, 2, 1, 1)
- (3, 2, 1, 1, 2)
- (3, 2, 1, 2, 1)
- (3, 2, 2, 1, 1)

Sample Input 2

5 3 4

Sample Output 2

204

Sample Input 3

998244 998353 998107

Sample Output 3

701926019